

AQI_METER

About :

Pluto drone is equipped with smoke sensor to assess a smokey environment and based on the analysis, manipulate the RGB colors on the Pluto drone PCB to imitate AQI categories

Add-On used :

The MQ-6 is a gas sensor used to detect smoke and flammable gases such as LPG, propane, and butane.

Components and Connection:

Pluto drone, MQ-6 and connecting wires.

Connect the MQ-6 sensor on the ADC port of Primus V5/X2 board

Primus V5/X2 pin	Peripherals
VCC on ADC port	VCC on MQ-6
GND on ADC port	GND on MQ-6
Signal on ADC port	A0 on MQ-6

How does it work?

The MQ-6 sensor provides range of values mainly from 500 to 1200 based on the amount of smoke.

So, for each category of smoke, we have divided it four different categories with the range of values. So accordingly the LED displays different colors.

Air type	(ADC_1_input)	Drone color	How to produce
Clean air	500 to 600	Green	Green ON
Light smoke	600 to 750	Yellow	Red and Green ON
Heavy smoke	7500 to 900	Pink	Red and Blue ON
Very dense smoke	900 to 1200	Red	Red ON

PlutoBlocks

Select Receiver Mode **Rx_ESP**

on start

- GREEN ON
- BLUE ON

pluto loop

- if (Read ADC Pin ADC_1 > 80 and Read ADC Pin ADC_1 < 800)
 - do
 - RED OFF
 - GREEN ON
 - BLUE OFF
- else if (Read ADC Pin ADC_1 > 800 and Read ADC Pin ADC_1 < 2000)
 - do
 - RED ON
 - GREEN ON
 - BLUE OFF
- else if (Read ADC Pin ADC_1 > 2000 and Read ADC Pin ADC_1 < 3700)
 - do
 - RED ON
 - GREEN OFF
 - BLUE ON
- else if (Read ADC Pin ADC_1 > 3700 and Read ADC Pin ADC_1 < 4500)
 - do
 - RED ON
 - GREEN OFF
 - BLUE OFF

LoopTime (ms) 100

on stop

Working:

Initially, we set receiver mode as ESP ,Green and Blue to be on.

Then when it detects the smoke based on ranges it turns on and off different types of colors imitating a AQI meter.

Different colors combine to form new ones, we have used this fact to make this AQI Meter.

Moving further

- Try adding actual color LEDs to Pluto, making it more appealing.
- Try flying multiple such drone in smokey environments and enjoy the light show you will have!!